Event ends in 1 day 4 hours 8 minutes.

Prerequisites 📋

Important
- Use Kiro **CLI** for this section
- Complete Setting Up Kiro

Situation 1: Network Troubleshooting 🔍

As a cloud engineer, you received a message from your tester asking for help accessing their EC2 instance in the VPC. They want to access the UI.

⚠️ The tester mentioned that the EC2 instance name is `load-generator` in us-east-1

Let's use Kiro CLI to help us 🚀

1. Within the VS Code Server environment, start Kiro CLI by running:

```
kiro-cli chat
```

2. Ask Kiro CLI to find the instance in question 🔎

```
Please help me find EC2 instances with names containing 'load-generator' in us-east-1
```

```
● Completed in 0.913s
I found 1 EC2 instance with 'load-generator' in its name in the us-east-1 region:

Instance details:
• Instance ID: i-087d8cdef3a3190e6
• Name: sampleapp-load-generator
• State: running
• Instance Type: t3.micro
• Public IP: 3.82.126.241
• Public DNS: ec2-3-82-126-241.compute-1.amazonaws.com
• Private IP: 10.0.1.86
• Private DNS: ip-10-0-1-86.ec2.internal
• Availability Zone: us-east-1a
• VPC ID: vpc-0bb7c3ff50d454d2f
• Subnet ID: subnet-0d6576030e2ffaff4
• AMI ID: ami-09e6f87a47903347c
• Launch Time: 2025-06-16T03:48:27+00:00 (today)
• Key Pair: sampleapp-key-pair

This instance appears to be part of a CloudFormation stack named "sample-app" and is tagged as a load generator instance.
```

3. Ask Kiro CLI to troubleshoot the issue 🛠️

```
I can't access the UI on port 80 for that EC2 instance.
Please help me find the root cause and suggest a fix. Also tell me about the possible impact of the fix.
Let me decide if I want to proceed.
```

```
● Completed in 0.894s
I've identified the issue preventing you from accessing the UI on the load generator EC2 instance. The problem is with the security group configuration.

## Root Cause:
The security group (sg-0659c4a6ad40c9823) has very restrictive inbound rules:

1. Limited IP Access: The security group only allows HTTP (port 80) and SSH (port 22) access from a single IP address: 49.49.242.24/32

2. Your IP Not Allowed: Your current IP address is not in the allowed list, which is why you cannot access the UI. The server is running and the web service is
active, but the security group is blocking your connection.
```

```
## Suggested Fix:
Update the security group to allow HTTP access from your current IP address:

aws ec2 authorize-security-group-ingress \
  --group-id sg-0659c4a6ad40c9823 \
  --protocol tcp \
  --port 80 \
  --cidr YOUR_IP_ADDRESS/32 \
  --region us-east-1


You would need to replace YOUR_IP_ADDRESS with your actual public IP address.

## Potential Impacts:

### Positive Impacts:
1. Immediate Access: You'll be able to access the UI on the load generator instance
2. Minimal Security Risk: By limiting access to only your IP address, you maintain a strong security posture
3. No Service Disruption: This change doesn't require restarting any services or the instance

### Negative Impacts:
1. IP Address Changes: If your IP address changes (due to reconnecting, using a different network, etc.), you'll lose access again
2. Additional Security Group Rule: Adds another entry to the security group, which might make management slightly more complex
3. Potential Compliance Issues: If there are specific compliance requirements that dictated the original IP restriction, this change might need approval
```

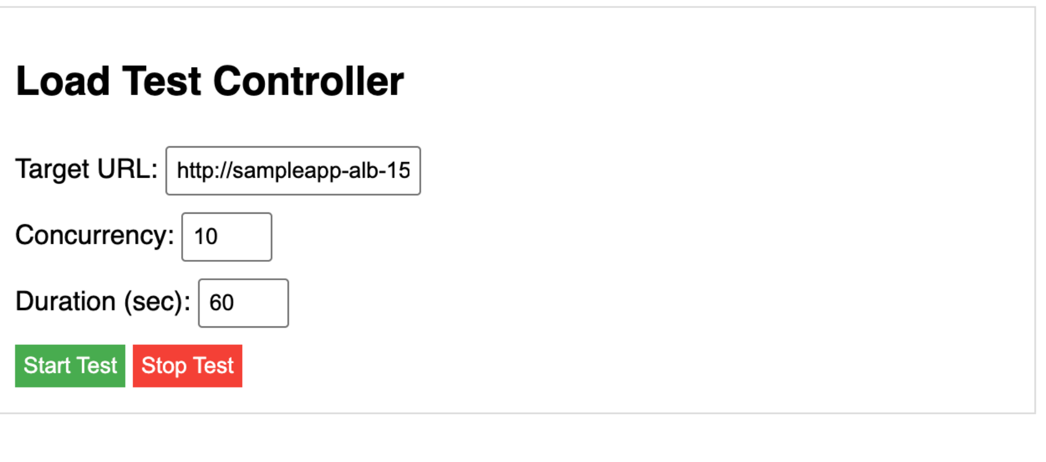
4. Now that you've found the root cause and fix, confirm with Kiro CLI to proceed with the fix ✅

```
proceed! make sure port 80 is accessible from 0.0.0.0/0
```

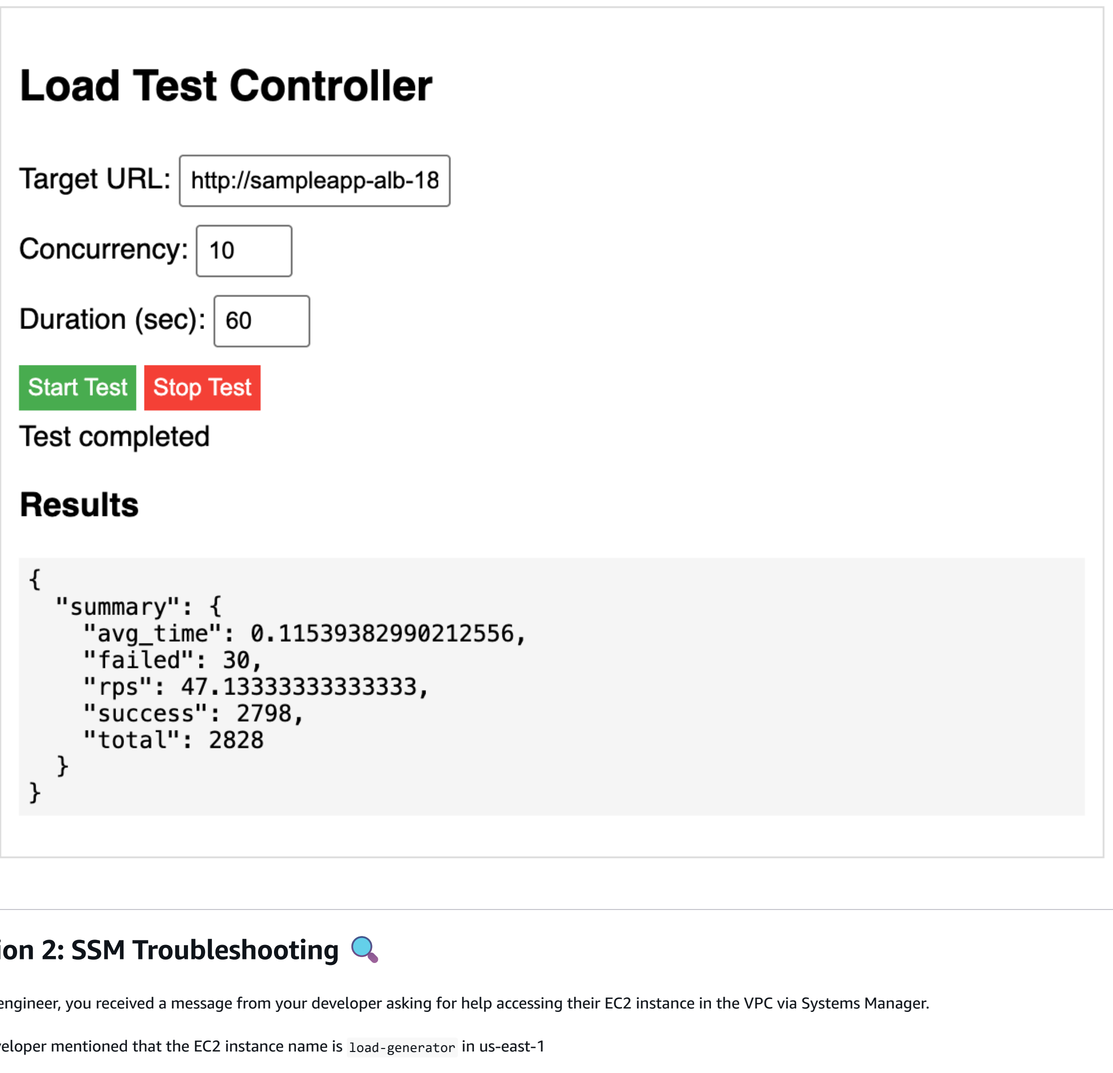
Important
⚠️ Ideally, you would want to allow access only from your own IP. However, in this lab, since Kiro CLI is running on VS Code, it assumes your IP is the VS Code server IP. You won't be able to access the load test server via your laptop because it will whitelist the VS Code server IP, not your laptop's public IP. So in this lab, we will ask Kiro CLI to allow all public IPs.

```
> 🎉 Success! The fix has been applied and verified.

## **What was changed:**
• Added inbound rule to security group sg-0f3cb775359c1f961
• **Protocol**: TCP
• **Port**: 80
• **Source**: 0.0.0.0/0 (Internet access)
• **Rule ID**: sgr-0326ef89fec28970c

## **Current Security Group Rules for Port 80:**
• ✅ 49.49.242.24/32 (original restricted access)
• ✅ 0.0.0.0/0 (new internet access)

## **Verification Results:**
• ✅ Port 80 is now accessible from the internet
• ✅ Connection test to 54.224.33.143:80 succeeded

## **Next Steps:**
You can now access the load generator UI at:
• **HTTP**: http://54.224.33.143
• **Public DNS**: http://ec2-54-224-33-143.compute-1.amazonaws.com

## **Security Reminder:**
The service is now publicly accessible. Consider:
1. Monitor access logs for unexpected traffic
2. Remove this rule when no longer needed using:
```

5. Try accessing the load test UI via HTTP 🌐



6. Try running the test using the default configuration ⚡



Situation 2: SSM Troubleshooting 🔍

As a cloud engineer, you received a message from your developer asking for help accessing their EC2 instance in the VPC via Systems Manager.

⚠️ The developer mentioned that the EC2 instance name is `load-generator` in us-east-1

Let's use Kiro CLI to help us 🚀

1. Within the VS Code Server environment, start Kiro CLI by running:

```
kiro-cli chat
```

2. Ask Kiro CLI to find the instance in question 🔎

```
Please help me find EC2 instances with names containing 'load-generator' in us-east-1
```

```
● Completed in 0.864s

> I found one EC2 instance with a name containing 'load-generator' in the us-east-1 region:

Instance Details:
• **Name**: sampleapp-load-generator
• **Instance ID**: i-0ccce7fcc0e919c1b
• **State**: running
• **Instance Type**: t3.micro
• **Launch Time**: 2025-07-24T07:53:27+00:00
• **Availability Zone**: us-east-1a
• **Public IP**: 54.224.33.143
• **Private IP**: 10.0.1.27
• **Public DNS**: ec2-54-224-33-143.compute-1.amazonaws.com
• **Key Pair**: sampleapp-key-pair
• **Security Group**: sample-app-LoadGeneratorSecurityGroup-3wiYZf9gaXuY
```

3. Ask Kiro CLI to troubleshoot the issue 🛠️

```
I can't connect to the instance using Systems Manager.
Please help me find the root cause and suggest a fix. Also tell me about the possible impact of the fix.
Let me decide if I want to proceed.
```

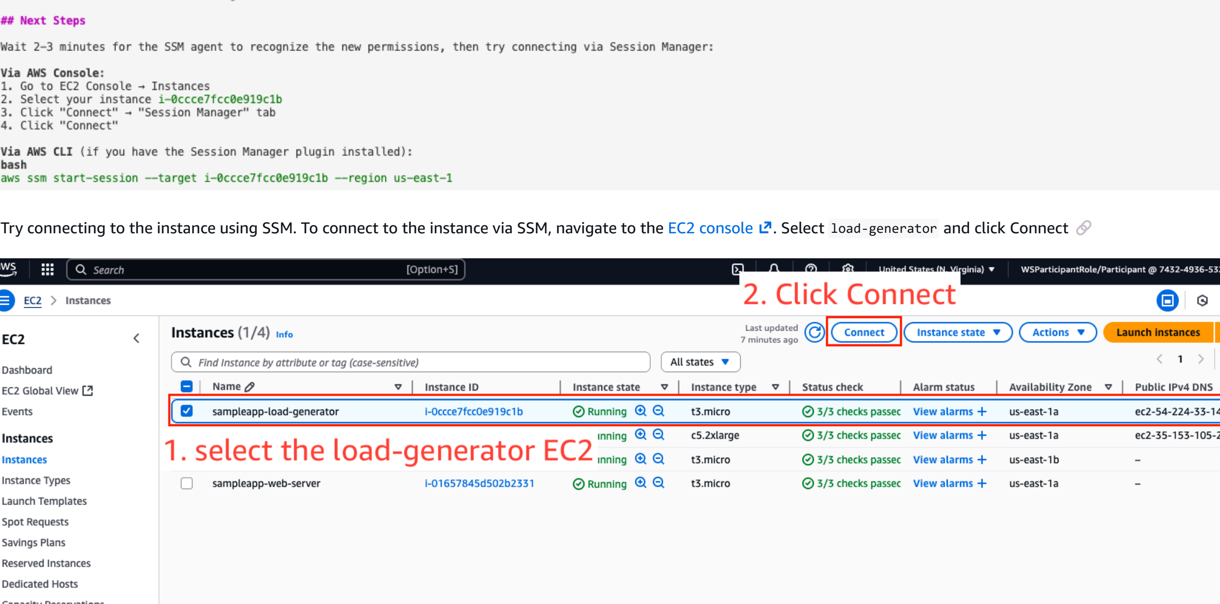
```
## Root Cause Analysis

Primary Issue: The instance lacks an IAM instance profile with the necessary permissions for Systems Manager Session Manager.

Current Status:
• ✅ Instance is registered with SSM and showing "Online"
• ✅ SSM Agent is installed and running (version 3.3.2299.0)
• ✅ Network connectivity exists (public subnet with internet gateway)
• ❌ No IAM instance profile attached to the instance
• ❌ No VPC endpoints for SSM (relying on internet connectivity)

## Suggested Fix

Create and attach an IAM role with the AmazonSSMManagedInstanceCore policy:

bash
# 1. Create IAM role
aws iam create-role --role-name SSMInstanceRole --assume-role-policy-document '{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Effect": "Allow",
      "Principal": {
        "Service": "ec2.amazonaws.com"
      },
      "Action": "sts:AssumeRole"
    }
  ]
}'

# 2. Attach the managed policy
aws iam attach-role-policy --role-name SSMInstanceRole --policy-arn arn:aws:iam::aws:policy/AmazonSSMManagedInstanceCore

# 3. Create instance profile
aws iam create-instance-profile --instance-profile-name SSMInstanceProfile

# 4. Add role to instance profile
aws iam add-role-to-instance-profile --instance-profile-name SSMInstanceProfile --role-name SSMInstanceRole

# 5. Associate with your instance
aws ec2 associate-iam-instance-profile --instance-id i-0ccce7fcc0e919c1b --iam-instance-profile Name=SSMInstanceProfile --region us-east-1


## Impact Assessment

Positive Impacts:
• ✅ Enables Session Manager connectivity
• ✅ Allows secure shell access without SSH keys
• ✅ Provides session logging and auditing capabilities
• ✅ No network changes required (uses existing internet connectivity)

Potential Risks:
• ⚠️Temporary disruption: Brief interruption during IAM profile association
• ⚠️Permission scope: The role grants broad SSM permissions (managed policy)
• ⚠️CloudFormation drift: Since this is a CloudFormation-managed resource, manual changes may cause drift

Considerations:
• The instance will need to be stopped/started or the SSM agent restarted after role attachment
• Since this is part of a CloudFormation stack, consider updating the template instead for long-term management
• Alternative: Create VPC endpoints for SSM to avoid internet dependency (more complex but more secure)
```

4. Now that you've found the root cause and fix, confirm with Kiro CLI to proceed with the fix ✅

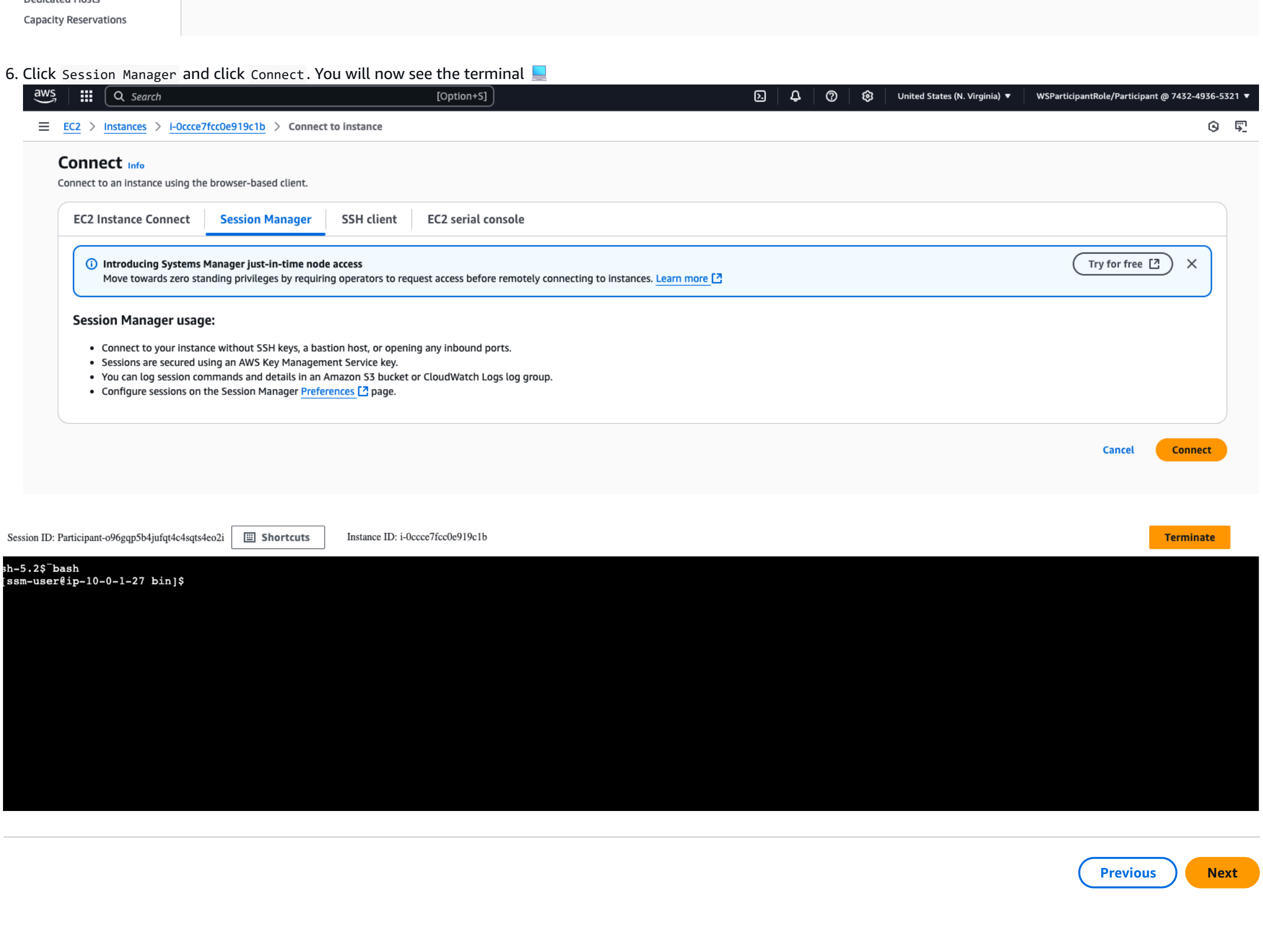
```
proceed! and restart the server
```

```
● Completed in 0.886s

> Perfect! The fix has been successfully implemented. Here's what I accomplished:

## ✅ Successfully Completed

1. Created IAM Role: SSMInstanceRole with EC2 trust policy
2. Attached Policy: AmazonSSMManagedInstanceCore managed policy
3. Created Instance Profile: SSMInstanceProfile
4. Associated Profile: Successfully attached to instance i-0ccce7fcc0e919c1b
5. Restarted Instance: Instance is now running with the new IAM profile

## Current Status

• **Instance State**: Running
• **IAM Instance Profile**: ✅ SSMInstanceProfile is now attached
• **IAM Role**: ✅ SSMInstanceRole with SSM permissions
• **Public IP**: 54.224.33.143 (unchanged)
• **Private IP**: 10.0.1.27 (unchanged)

## Next Steps

Wait 2-3 minutes for the SSM agent to recognize the new permissions, then try connecting via Session Manager:

Via AWS Console:
1. Go to EC2 Console → Instances
2. Select your instance i-0ccce7fcc0e919c1b
3. Click "Connect" → "Session Manager" tab
4. Click "Connect"

Via AWS CLI (if you have the Session Manager plugin installed):
bash
aws ssm start-session --target i-0ccce7fcc0e919c1b --region us-east-1
```

5. Try connecting to the instance using SSM. To connect to the instance via SSM, navigate to the EC2 console. Select `load-generator` and click Connect 🔗



6. Click `Session Manager` and click `Connect`. You will now see the terminal 💻



Previous
Next